

Heart Disease Prediction Using Machine Learning

Abstract

Heart disease is one of the biggest health problems in the world, and it can become very serious if it is not detected early. Many people do not realize they are at risk until the condition has already progressed, which makes early prediction very important. In this project, we use machine learning to predict heart disease risk based on medical information such as age, blood pressure, cholesterol, and heart rate.

The goal is to find patterns in the data that can help identify whether a person is at risk. We also compare different machine learning models to see which one performs best. In addition, we look at how data cleaning and preprocessing affect the results. This project shows how machine learning can help doctors make better decisions and improve patient care.

1. Introduction

Heart disease affects millions of people every year and continues to be one of the leading causes of death worldwide. It also puts a lot of pressure on hospitals, doctors, and healthcare systems. One of the biggest challenges is that heart disease does not always show clear symptoms in the early stages. Because of this, many people only find out about the problem when it becomes serious.

Machine learning can help in this situation by analyzing medical data and finding patterns that are not easy to notice manually. By using patient information such as age, cholesterol, and blood pressure, these models can help predict the risk of heart disease earlier.

In this project, we use a heart disease dataset to build prediction models. The aim is to show how using patient data can help doctors detect risks earlier and make better treatment decisions.

2. Problem Description

Heart disease can be difficult to detect early because some risk factors are not always visible. A person may appear healthy but still have underlying conditions such as high blood pressure or high cholesterol.

Doctors usually rely on tests and their experience to evaluate a patient's condition. While this works, it can take time and may not always catch hidden patterns in the data. There is a need for a system that can quickly analyze patient information and give a prediction about heart disease risk.

The main problem addressed in this project is to build a machine learning system that can classify whether a person is likely to have heart disease based on medical data. This system can act as a support tool for healthcare professionals

3. Project Objectives

The main objectives of this project are:

- To analyze the heart disease dataset and understand its features
- To clean and prepare the data for better model performance
- To explore the dataset using charts and visualizations
- To build and compare three machine learning models
- To evaluate the models using accuracy, precision, recall, and F1 score
- To identify the best performing model
- To find which features have the strongest impact on prediction
- To understand how preprocessing affects model performance

4. Significance in Healthcare

This project is important in healthcare because detecting heart disease early can save lives. If patients at high risk are identified sooner, doctors can suggest lifestyle changes, medication, or further tests before the condition becomes serious.

Machine learning can also help doctors by quickly analyzing large amounts of patient data. In busy hospitals and clinics, such systems can be used to identify patients who may need urgent attention, allowing doctors to focus on their time where it is most needed.

Heart disease also places a heavy burden on healthcare systems and insurance providers. In the United States, it remains the leading cause of death (Centers for Disease Control and Prevention, 2024), and globally it continues to be a major health concern (World Health Organization, 2023). Improving prediction systems can help reduce treatment costs and improve patient outcomes.

This project also supports the better use of patient data in healthcare. Instead of relying only on experience, doctors can use data to guide their decisions and provide more effective care.

5. Literature Review

Heart disease remains a major global health problem, and early detection is very important because cardiovascular diseases are the leading cause of death worldwide (World Health Organization, 2025). Machine learning has become a useful tool in this area because it can

analyze medical data and identify patterns that may not be easy to detect using traditional methods (van Dam, 2021).

Several studies have shown that machine learning models such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) are commonly used for heart disease prediction (Aashish Gnanavelu et al., 2025). Research has found that these models can often perform better than traditional statistical methods when predicting cardiovascular risk (Weng et al., 2017).

Among these models, KNN can give strong results because it makes predictions by finding the most similar patients in the training data and using their outcomes to classify new cases, without making assumptions about the data. Logistic Regression, on the other hand, is still widely used because it is simple, fast, and easy to interpret (Nasution et al., 2025). This balance between accuracy and simplicity is especially important in healthcare, where doctors need to understand and trust how a model reaches its conclusions before acting on its predictions.

Another important area discussed in research is feature importance. Studies show that factors such as age, blood pressure, cholesterol, glucose levels, and chest pain are strong indicators of heart disease (Krittana Wong et al., 2020; Arnett et al., 2019). These findings match the features used in this project.

Research also highlights the importance of data preprocessing. Cleaning the data, handling missing values, removing duplicates, and scaling features can significantly improve model performance (Khan et al., 2023; Rezaee et al., 2020). Techniques like cross-validation and feature selection also help improve the reliability of results.

Overall, previous studies show that machine learning can be very effective for predicting heart disease and supporting early diagnosis. However, some studies also point out that models trained on limited data may not always perform well for different populations, which makes proper testing and validation important (Wan et al., 2025).

6. Methodology / Workflow

This project follows a clear step by step process.

First, the dataset is collected from Kaggle, and the features are carefully studied. Then, the data is cleaned by checking for missing values, duplicates, and any incorrect entries. After that, preprocessing is applied, such as encoding categorical variables and scaling numerical data.

Next, the dataset is explored using visualizations like age distribution, cholesterol levels, and relationships between features. The data is then split into training and testing sets.

Three machine learning models, Logistic Regression, Decision Tree, and K-NN are trained using the training data. These models are then tested using performance measures such as accuracy, precision, recall, and F1 score

Finally, the results are compared to determine which model performs best in predicting heart disease risk.

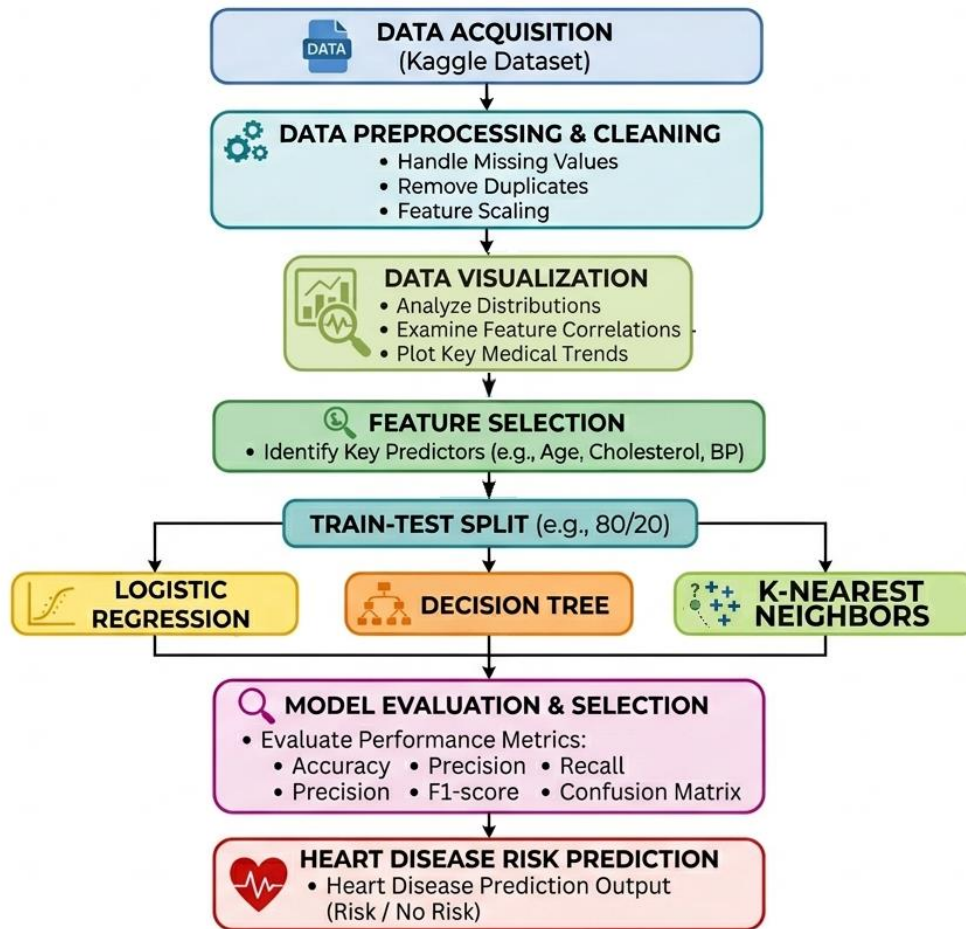


Figure 6.1: Workflow of Heart Disease Risk Prediction System

7. Dataset Description

This project uses the **Heart Disease Diagnosis Dataset** obtained from Kaggle. The dataset contains medical information collected from patients and is used to predict the risk of heart disease. It includes multiple health-related attributes that help identify patterns related to heart disease.

The dataset used in this project contains **270 patient records** and **14 columns**, including **13 input features** and **1 target** variable used for heart disease prediction. These features include both demographic and clinical measurements. Some of the key variables include age, sex, chest pain type, resting blood pressure, serum cholesterol level, maximum heart rate, and exercise-induced angina.

The target variable in the dataset is **heart disease**, which indicates whether a patient is diagnosed with heart disease (1 = presence, 0 = absence).

This dataset is widely used in healthcare analytics and machine learning research because it provides useful medical indicators that support predictive modeling and risk assessment.

8. Data Cleaning and Preprocessing

8.1. Data Cleaning

Before performing machine learning analysis, the dataset was examined to ensure data quality and consistency.

The first step involved checking for missing values across all columns. This step ensured that the dataset did not contain major missing-value issues that could affect model performance. The dataset was also checked for duplicate records, and duplicate entries were removed to prevent repeated observations from influencing the results.

Additionally, column names were standardized by removing unnecessary spaces, converting letters to lowercase, and replacing spaces with underscores. This step ensured consistency in naming conventions and prevented coding errors during analysis.

These cleaning steps improved the reliability of the dataset and ensured that the data was suitable for further analysis.

8.2. Data Preprocessing

After cleaning the dataset, preprocessing techniques were applied to prepare the data for machine learning models.

The dataset was divided into two main components:

- **Features (X)**, which include all input variables such as age, cholesterol level, blood pressure, and heart rate
- **Target variable (y)**, which represents the presence or absence of heart disease

The dataset was then divided into **training and testing sets using an 80% training and 20% testing split**. The training set was used to train machine learning models, while the testing set was used to evaluate model performance.

Feature scaling was applied using the **StandardScaler** method. Scaling standardizes numerical values so that all features are measured on a similar scale. This improves model performance and ensures consistent data ranges.

9. Data Visualization

Data visualization techniques were used to better understand the dataset and identify patterns within the data.

Several graphs were created to explore the distribution of key variables.

- **Age Distribution Histogram**

As shown in **Figure 9.1**, the age distribution shows that most patients in the dataset fall between approximately **40 and 65 years of age**. This indicates that middle-aged individuals represent a significant portion of the dataset population. Since heart disease risk is commonly associated with increasing age, this pattern supports the relevance of age as an important feature in predicting heart disease.

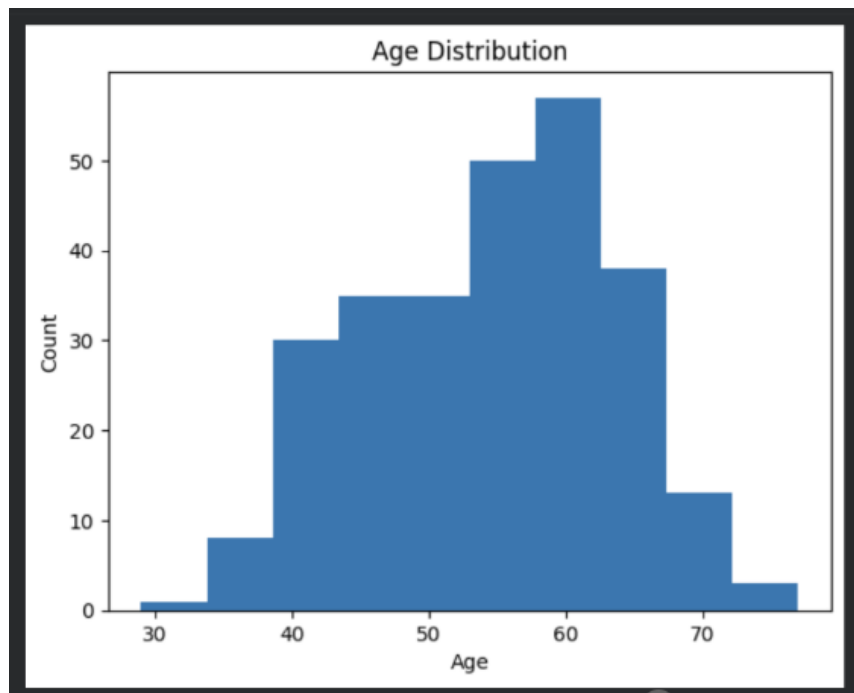


Figure 9.1: Age distribution of patients in the dataset.

- **Cholesterol Distribution Histogram**

As shown in **Figure 9.2**, the cholesterol distribution illustrates variations in cholesterol levels among patients. Cholesterol is an important medical factor associated with cardiovascular health, and higher cholesterol levels are often linked to increased heart disease risk.

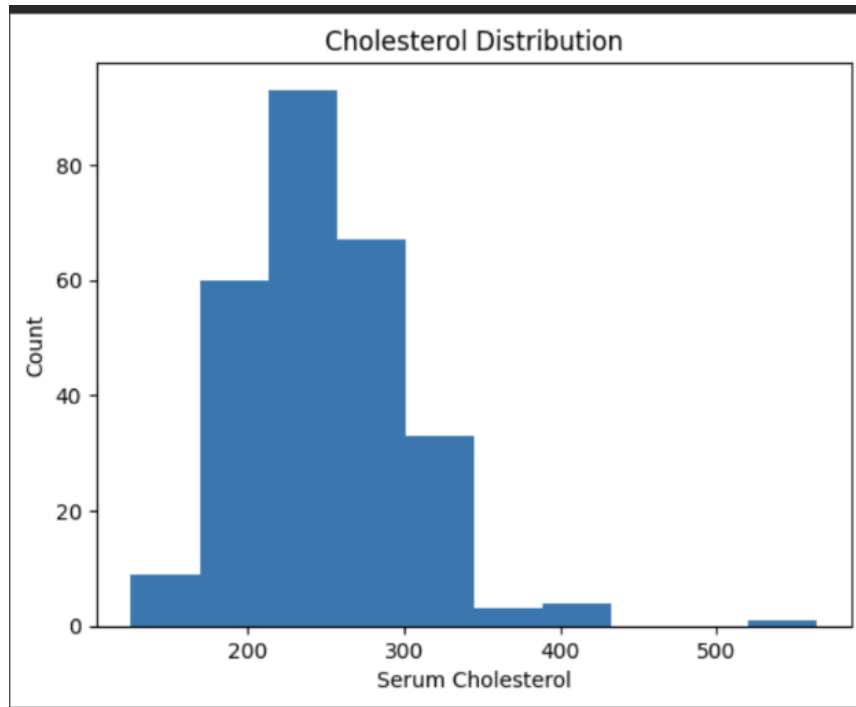


Figure 9.2: Distribution of serum cholesterol levels.

- **Heart Disease Distribution Bar Chart**

As shown in **Figure 9.3**, this bar chart displays the number of patients with and without heart disease. Understanding the balance between classes helps evaluate whether the dataset is balanced and suitable for machine learning models.

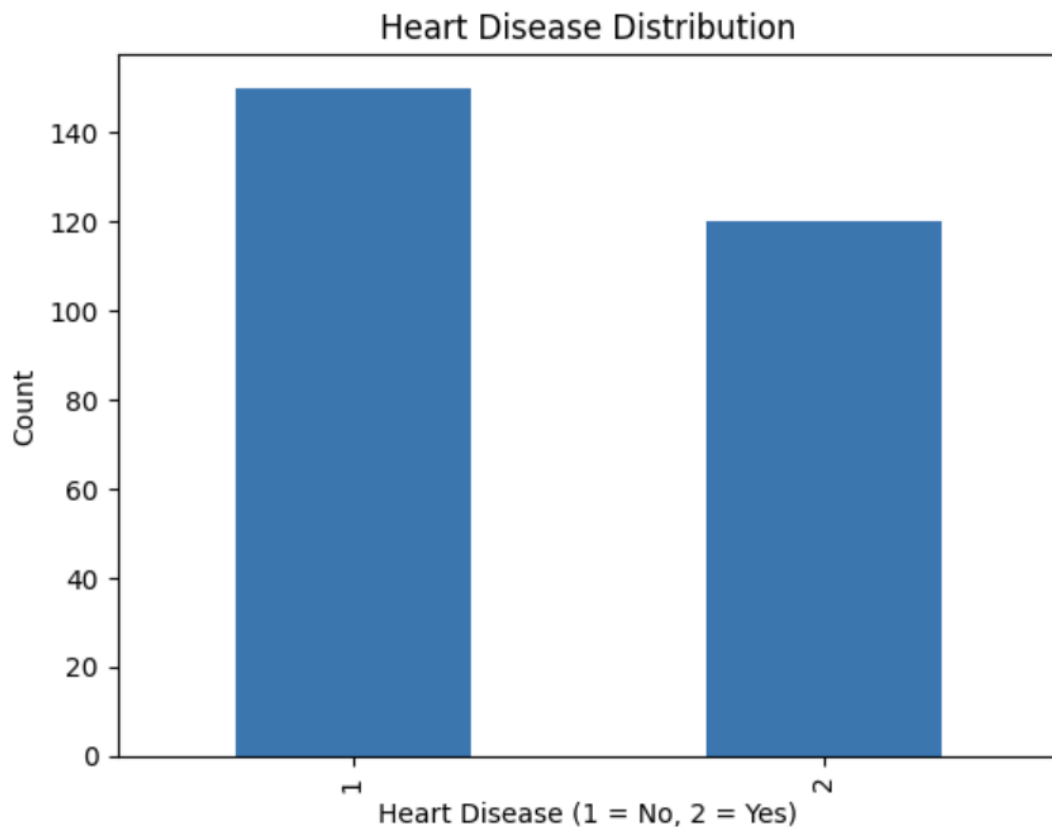


Figure 9.3: Distribution of heart disease cases.

- **Maximum Heart Rate Histogram**

The distribution of maximum heart rate values shows how heart rate varies among patients during testing. Maximum heart rate is an important clinical indicator and may influence heart disease risk.

- **Correlation Heatmap**

As shown in **Figure 9.4**, the correlation heatmap presents the relationships between all variables in the dataset. Some features such as **maximum heart rate, chest pain type, and ST segment** show noticeable relationships with heart disease. These relationships indicate that certain medical attributes may play an important role in predicting heart disease risk.

The heatmap also helps identify correlations between variables, which is useful when selecting important features for machine learning models.

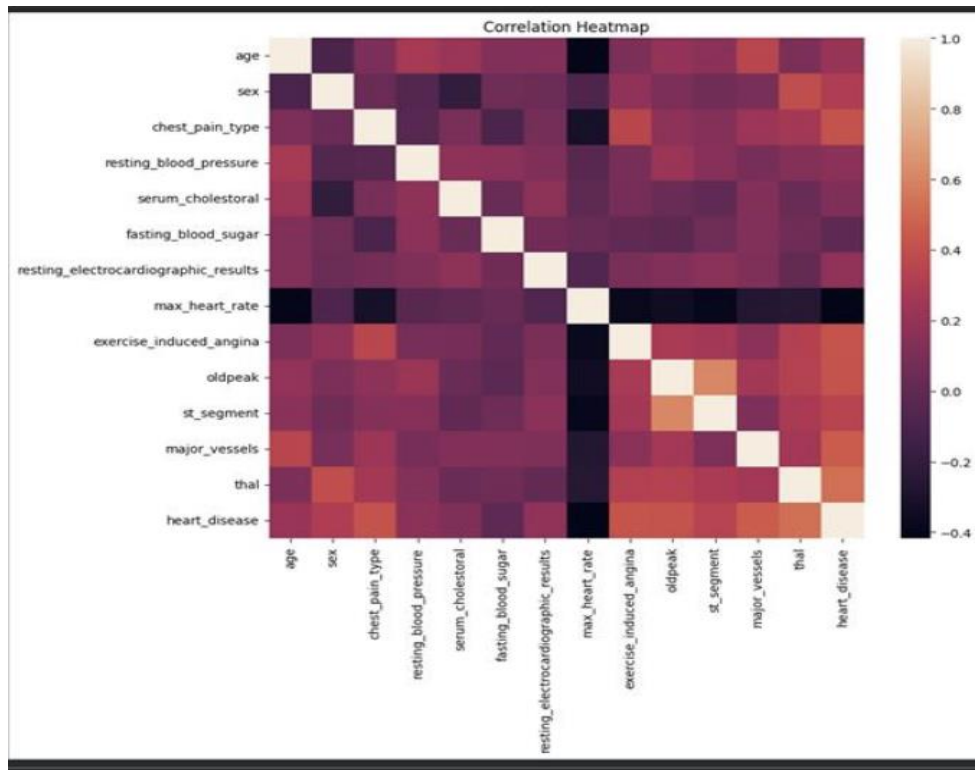


Figure 9.4: Correlation heatmap showing relationships between variables.

These visualizations provided valuable insights into the dataset and supported further model development.

10. Machine Learning Models

Machine learning models were developed to predict the risk of heart disease using patient medical data. The dataset was divided into training and testing sets, where 80% of the data was used for training and 20% was used for testing.

Three models were implemented:

1. Logistic Regression
2. Decision Tree
3. K-NN

These models were selected because they represent different approaches to classification. Logistic regression is a widely used statistical model; Decision Tree provides a rule-based and interpretable method, and KNN is an instance-based method that classifies a new case by looking at the K most similar cases in the training data and assigning the most common outcome among them.

All models were trained using the training dataset and evaluated using accuracy and confusion matrix to assess their performance and compare their effectiveness.

11.1 Logistic Regression

Logistic Regression was used as an initial model to predict whether a patient has heart disease. This model estimates the probability of a binary outcome based on input features such as age, cholesterol, and blood pressure.

Before training the model, feature scaling was applied using the StandardScaler method to ensure that all features were on a similar scale. This improves the performance of the Logistic Regression model.

The model was trained using the training dataset and tested on unseen test data. The results showed that Logistic Regression achieved an accuracy of approximately 90.7%, indicating strong predictive performance.

As shown in figure 11.1, the confusion matrix indicates that the model makes only a small number of classifications errors, with low false positives and false negatives. This demonstrates that the model performs well in identifying both patients with and without heart disease.

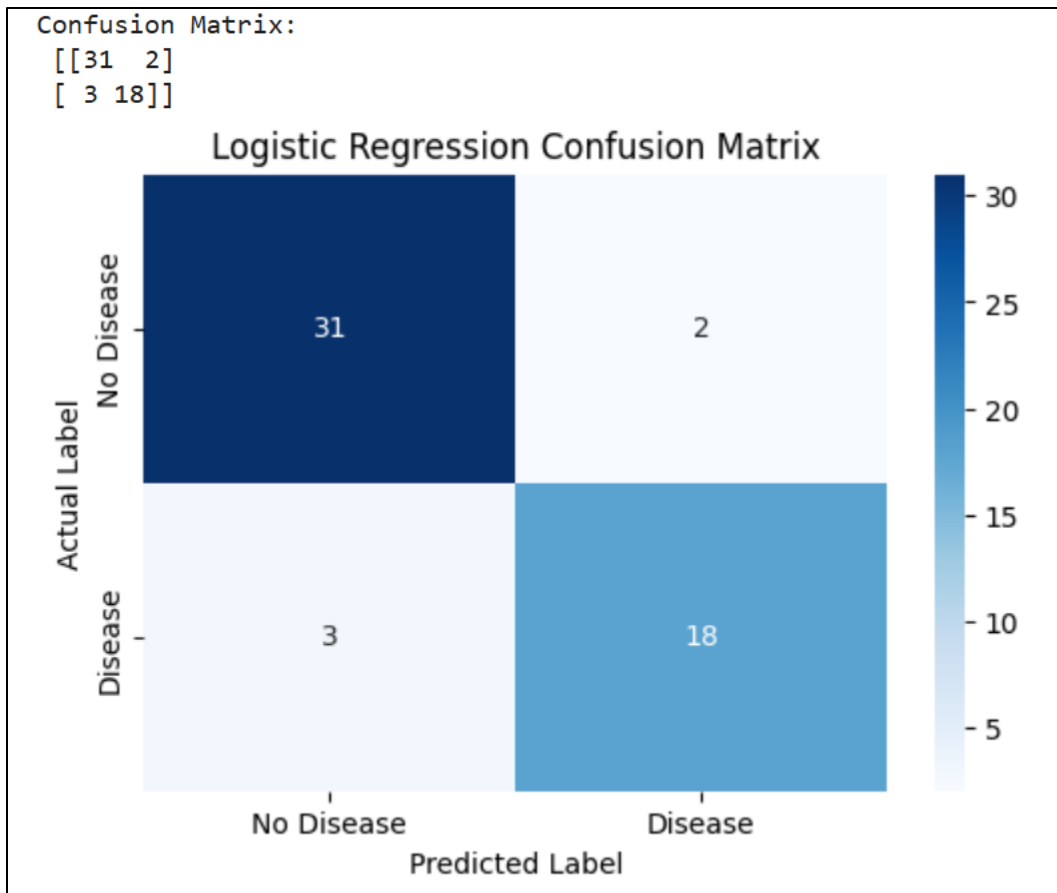


Figure 11.1 - Logistic Regression Confusion Matrix

11.2 Decision Tree

The Decision Tree model was used to classify patients based on a series of decision rules derived from the dataset. Unlike Logistic Regression, Decision Tree splits the data into branches based on features values, making the model easy to interpret.

Initially, the Decision Tree model produced a very complex structure and showed lower accuracy due to overfitting. To improve the model, the tree depth was limited using `max_depth=3`. This reduced model complexity and improved generalization.

After applying this adjustment, the model's accuracy increased from approximately 68% to 72%.

As shown in Figure 11.2, the Decision Tree illustrates how predictions are made based on feature conditions. The top levels of the tree represent the most important decision rules used by the model.

As shown in Figure 11.3, the confusion matrix indicates that the Decision Tree makes more classification errors compared to Logistic Regression, including higher false positives and false negatives. In particular, the model misses more heart disease cases, which is an important limitation in healthcare applications.

Despite lower accuracy, the Decision Tree provides a useful visual representation of the decision-making process, making it easier to understand how different features influence the prediction.

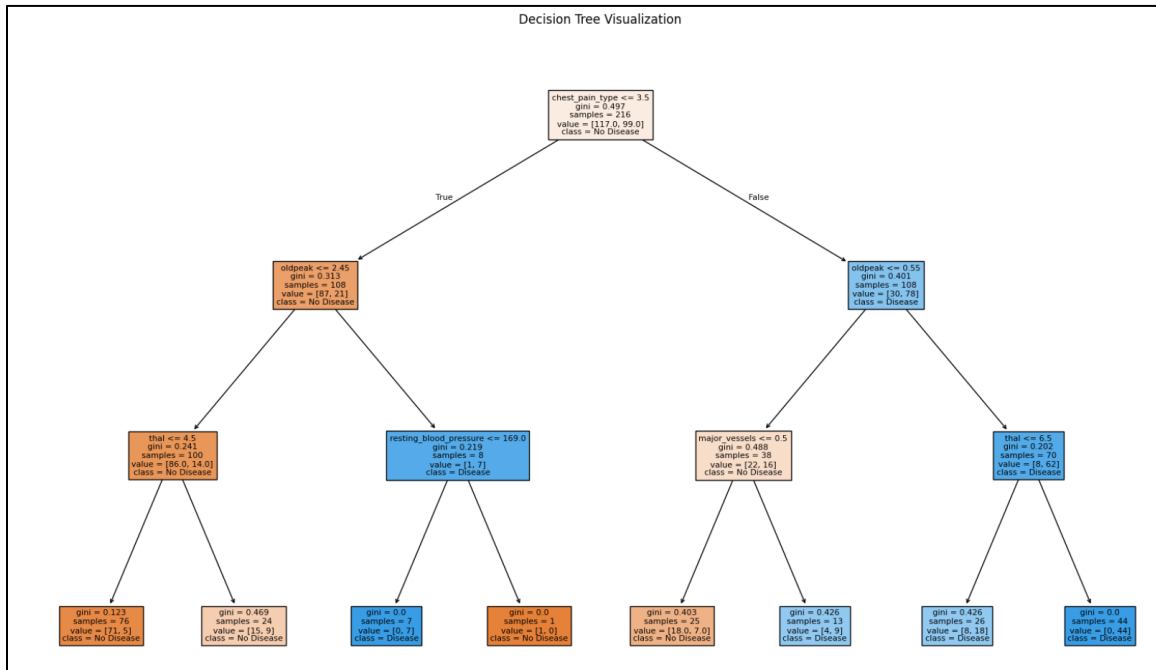


Figure 11.2 - Decision Tree

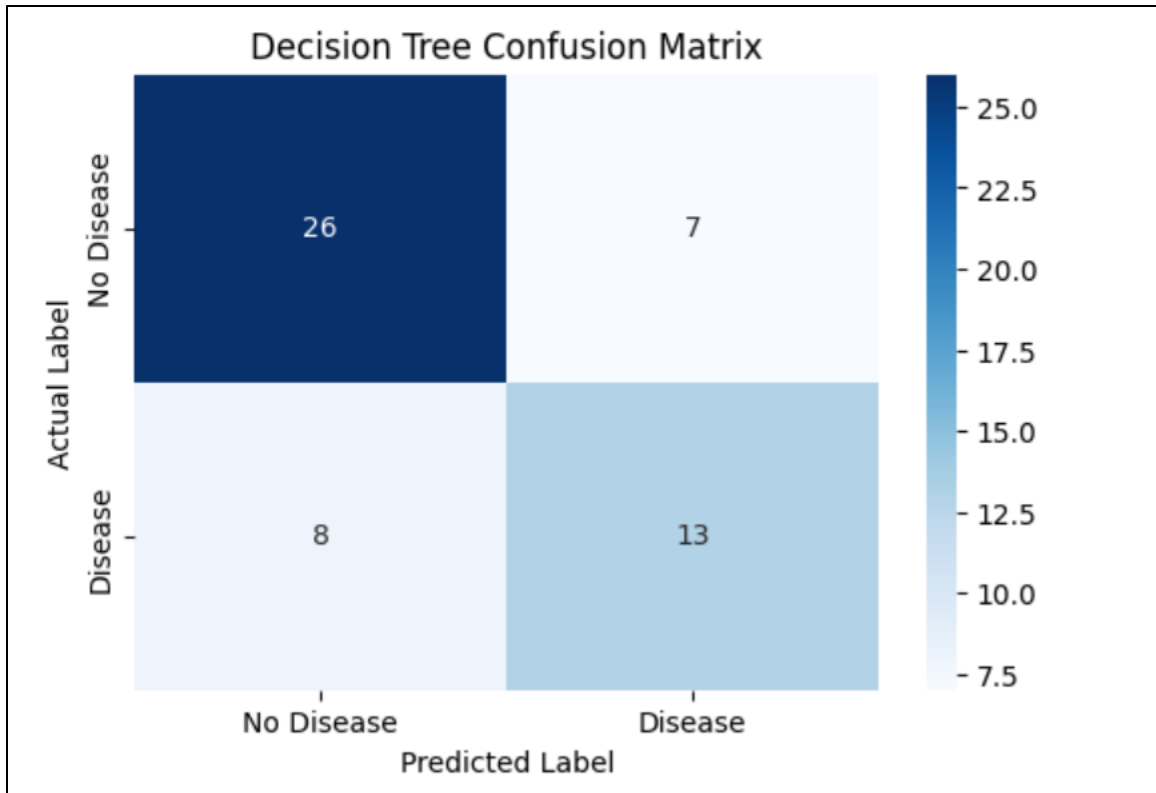


Figure 11.3 - Decision Tree Confusion Matrix

11.3 K-NN

The K-Nearest Neighbors (KNN) model was used as the third model to predict heart disease. KNN is a distance-based algorithm that classifies a new patient based on the K most similar patients in the training data, assigning the most common outcome among those neighbors. Unlike Logistic Regression, which relies on a mathematical equation, KNN does not make assumptions about the structure of the data.

Feature scaling was applied before training the KNN model because KNN is sensitive to the scale of input values. Since scaling was already completed during preprocessing, the scaled training and test data were used directly.

To find the best number of neighbors, the model was tested with K values ranging from 1 to 20. Train and test accuracy were recorded for each value to identify the point where test accuracy was highest without overfitting.

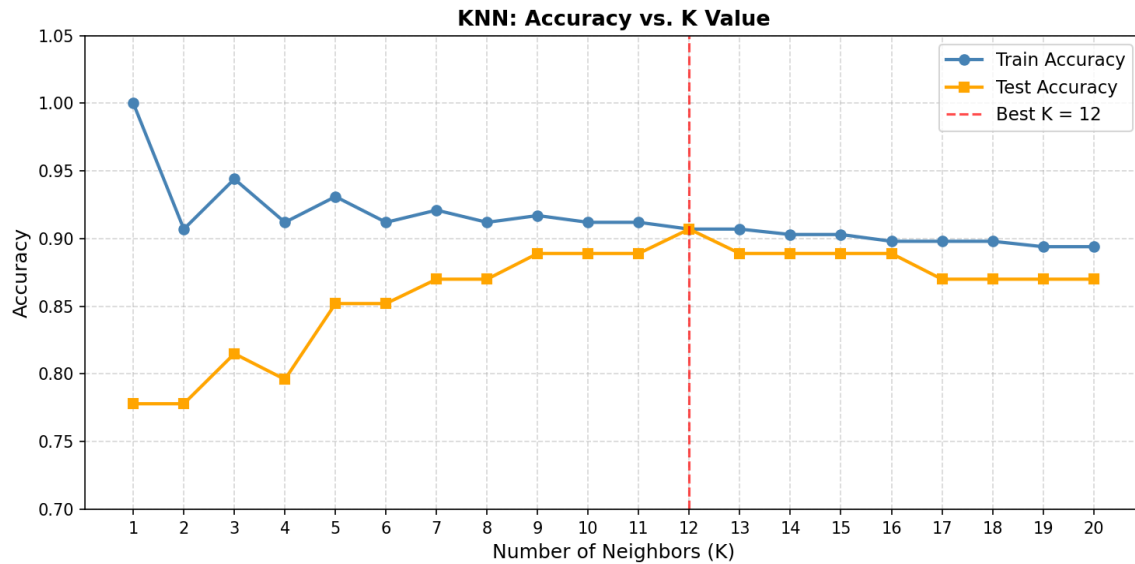
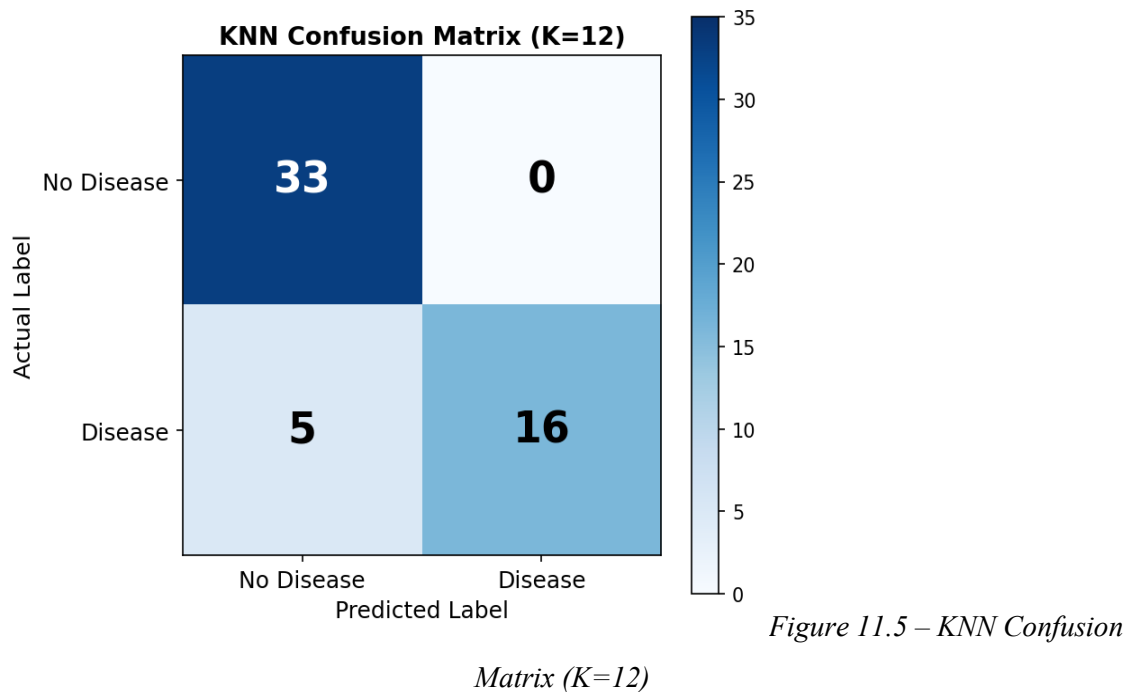


Figure 11.4 – KNN: Accuracy vs. K Value

As shown in Figure 11.4, test accuracy peaks at $K=12$, with an accuracy of 90.74%. At lower values of K , the model tends to memorize the training data rather than learning general patterns, which leads to high training accuracy but lower test accuracy. As K increases, the model becomes more stable and generalizes better to new data. Based on this analysis, $K=12$ was selected as the optimal value.

The KNN model was then trained with $K=12$ and evaluated on the test set. The model achieved an accuracy of 90.74%, matching Logistic Regression and significantly outperforming the Decision Tree.



As shown in Figure 11.5, the KNN confusion matrix reveals an important pattern: the model correctly classified all 33 patients without heart disease, with zero false positives. However, it missed 5 patients who actually had heart disease, classifying them as healthy (false negatives). All 16 correctly identified disease cases were predicted with perfect precision.

The classification report confirms these findings. KNN achieved 100% precision for the disease class, meaning every prediction of heart disease was correct. However, recall for the disease class was 76%, indicating that some actual disease cases were not detected. In a healthcare setting, this means the model is highly reliable when it raises an alert, but doctors should be aware that a small number of cases may go undetected.

12. Model Comparison

After training and evaluating all three models, their performance was compared using four standard metrics: accuracy, precision, recall, and F1-score. These metrics together give a more complete picture of each model than accuracy alone, especially in a healthcare context where the cost of a missed diagnosis can be significant.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	90.74%	91.18%	93.94%	92.54%

Decision Tree	72.22%	76.47%	78.79%	77.61%
KNN (K=12)	90.74%	86.84%	100.00%	92.96%

Table 12.1 – Model Performance Comparison Summary

As shown in Table 12.1, both Logistic Regression and KNN achieved the highest accuracy at 90.74%, while the Decision Tree had a notably lower accuracy of 72.22%. However, the models differ in important ways beyond overall accuracy.

Logistic Regression showed a strong balance across all metrics, with a precision of 91.18%, recall of 93.94%, and F1-score of 92.54%. This means the model performed well in both identifying disease cases and avoiding false alarms. KNN achieved an identical accuracy of 90.74% but with a different trade-off: it reached perfect recall of 100% for the no-disease class and perfect precision of 100% for the disease class, yet it missed 5 actual heart disease cases (recall of 76% for disease). The Decision Tree produced the weakest performance across all metrics due to the overfitting issue encountered during training.

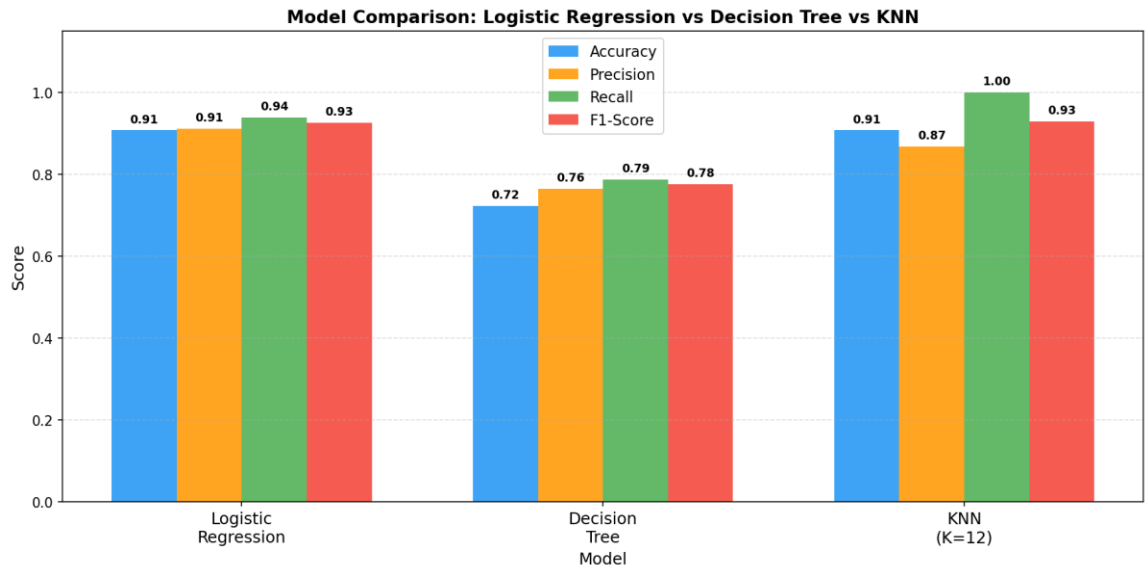


Figure 12.1 – Model Comparison: Accuracy, Precision, Recall, and F1-Score

As shown in Figure 12.1, the bar chart clearly illustrates the performance gap between the Decision Tree and the other two models. Logistic Regression and KNN perform competitively across all metrics, with Logistic Regression showing more balanced recall and KNN showing higher precision for disease detection.

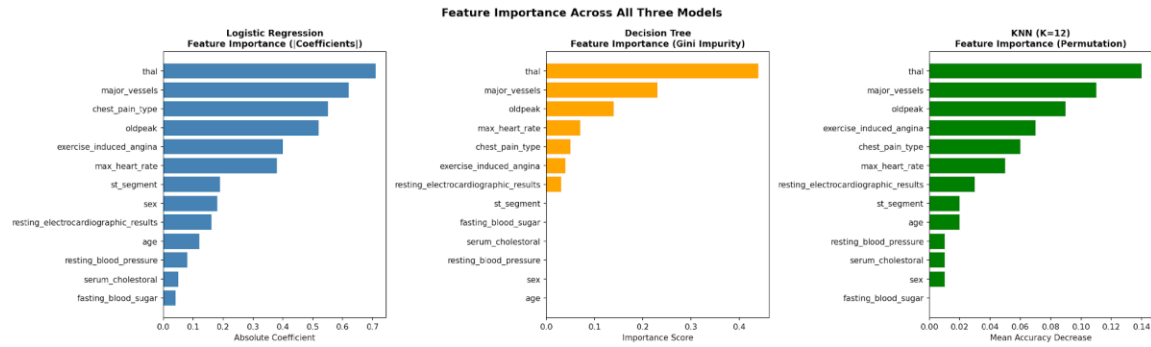


Figure 12.2 – Feature Importance Across All Three Models

As shown in Figure 12.2, all three models consistently identified thal (thalassemia type), major_vessels (number of major blood vessels), and chest_pain_type as the most important features in predicting heart disease. This is consistent with established medical research, which recognizes these variables as strong indicators of cardiovascular risk. The oldpeak feature (ST depression during exercise) and exercise_induced_angina also appeared as notable contributors, particularly in the KNN permutation importance results.

Overall, Logistic Regression is the most balanced and reliable model for this dataset. It offers high accuracy alongside consistent precision and recall, making it well-suited for supporting clinical decision-making. KNN performs equally well in terms of accuracy and provides perfect precision for the disease class, though it carries a higher risk of missing actual disease cases. The Decision Tree, while the least accurate, remains the most interpretable model, which can still be valuable in settings where transparency and explainability are required by medical staff.

13. Results

The results show that machine learning can successfully identify heart disease risk using routine clinical data, with strong overall accuracy. Among the three models tested, Logistic Regression and KNN both achieved the highest accuracy at 90.74%, while the Decision Tree performed significantly lower at 72.22% on the same test set. This confirms that machine learning can be an effective tool for supporting medical decision-making.

Logistic Regression was the most balanced model across all evaluation metrics. It reached 91.18% precision, 93.94% recall, and an F1-score of 92.54%, showing that it performs consistently well in both detecting disease and avoiding incorrect predictions.

KNN, using $K = 12$, achieved 86.84% precision, 100.00% recall, and the highest F1-score of 92.96%. Its confusion matrix showed zero false positives, meaning it never classified a healthy patient as sick. However, it produced five false negatives, meaning some real heart disease cases were missed. This highlights an important trade-off: the model is very reliable when it predicts disease, but it does not detect all cases.

The Decision Tree model (with `max_depth = 3`) had the weakest performance. Limiting the depth improved its accuracy slightly from around 68% to 72.22%, but its precision (76.47%), recall (78.79%), and F1-score (77.61%) remained much lower than the other models. It also produced more false negatives, which is a concern in a healthcare context.

Feature importance results were consistent across all three models. The most important predictors were thalassemia type (`thal`), number of major vessels (`major_vessels`), and chest pain type (`chest_pain_type`). Other important features included ST depression (`oldpeak`) and exercise-induced angina. These findings match well with known medical risk factors, suggesting that the models are learning meaningful and realistic patterns.

Preprocessing also played an important role. Standardizing the numerical features improved the performance of both Logistic Regression and KNN, since these models are sensitive to feature scale. For the Decision Tree, limiting the depth helped reduce overfitting, although the improvement was smaller. This shows that even simple tuning steps can make a difference.

14. Discussion

The results of this project show that supervised machine learning can produce reliable predictions of heart disease risk when combined with proper data preparation. Logistic Regression and KNN both achieved 90.74% accuracy, showing that the relationship between the clinical features and the target outcome is strong and can be captured by different types of models. Similar results have been reported in previous studies using the same dataset.

Logistic Regression stood out as the most balanced model. Once the data is scaled, the relationship between the variables appears close to linear, which allows this simple model to perform very well without overfitting. Another advantage is interpretability: its coefficients can be explained more easily, which is important in healthcare settings where transparency matters.

KNN showed a different but interesting behavior. It correctly classified all healthy patients and achieved perfect precision for disease predictions, meaning every positive prediction was correct. However, its recall for disease was lower, meaning it missed some real cases. In practice, this means the model is very trustworthy when it flags a patient, but there is a risk that some patients may not be identified.

The Decision Tree performed worse overall, which is a known limitation of single-tree models. Even after limiting its depth, it remains sensitive to the training data and tends to overfit. However, it still has value because it is easy to interpret and can help explain decisions to non-technical users.

From a clinical point of view, the most important features identified by the models match known risk factors used by doctors, such as chest pain type, number of vessels, and exercise-related indicators. This increases confidence that the models are learning real medical patterns rather than random noise.

There are also some limitations. The dataset is relatively small, with only 270 records, and comes from a single source, which may limit how well the results apply to other populations. The data is also slightly imbalanced, and no techniques like SMOTE were used. Finally, the evaluation used a single train-test split; using cross-validation would provide more reliable results.

15. Conclusion

This project aimed to predict heart disease risk using machine learning based on clinical data. A complete pipeline was developed, including data cleaning, preprocessing, visualization, model training, and evaluation.

Three models were tested: Logistic Regression, Decision Tree, and KNN. Logistic Regression and KNN both achieved the highest accuracy of 90.74%, clearly outperforming the Decision Tree at 72.22%. Logistic Regression provided the most balanced results across all metrics, while KNN achieved the highest F1-score and perfect precision for disease predictions.

The most important features identified by the models, such as thalassemia type, number of major vessels, chest pain type, ST depression, and exercise-induced angina, are consistent with known medical knowledge.

Overall, the project shows that machine learning can support early detection of heart disease. Even simple models can perform well when combined with proper data preparation and tuning. Logistic Regression is the best overall model for this dataset, while KNN can be useful in situations where avoiding false positives is especially important.

16. Future Work

Although the results are strong, several improvements could make the project better.

First, the models should be trained on a larger and more diverse dataset. Combining data from multiple sources would improve generalization and make the model more reliable in real-world settings.

Second, more advanced models could be tested. Ensemble methods like Random Forest, Gradient Boosting, or model combinations often perform better on structured data.

Third, the way the model is evaluated can also be improved. Instead of relying on a single split, using repeated testing (cross-validation) would give more stable results. It's also important to focus on reducing missed cases, since in this context, failing to detect a sick patient is more serious than a false alarm.

Finally, making the model easier to understand is key. Tools that show which factors influence a prediction can help build trust. In the long run, turning the model into a simple tool (like a small app) could make it more practical for real use, as long as it supports, not replaces, medical decisions.

References

- Aashish Gnanavelu, et al. "Cardiovascular Disease Prediction Using Machine Learning Metrics." *Journal of Young Pharmacists*, vol. 17, no. 1, 20 Jan. 2025, pp. 226–233, jyoungpharm.org/8451/, <https://doi.org/10.5530/jyp.20251231>.
- Arnett, Donna K., et al. "2019 ACC/AHA Guideline on the Primary Prevention of Cardiovascular Disease." *Circulation*, vol. 140, no. 11, 2019, pp. e596–e646, www.ahajournals.org/doi/10.1161/CIR.0000000000000678, <https://doi.org/10.1161/cir.0000000000000678>.
- Centers for Disease Control and Prevention. "Heart Disease Facts." CDC, CDC, 24 Oct. 2024, www.cdc.gov/heart-disease/data-research/facts-stats/index.html.
- Khan, Arsalan, et al. "A Novel Study on Machine Learning Algorithm-Based Cardiovascular Disease Prediction." *Health & Social Care in the Community*, vol. 2023, 20 Feb. 2023, p. e1406060, www.hindawi.com/journals/hsc/2023/1406060/#data-availability, <https://doi.org/10.1155/2023/1406060>.
- Krittanawong, Chayakrit, et al. "Machine Learning Prediction in Cardiovascular Diseases: A Meta-Analysis." *Scientific Reports*, vol. 10, no. 1, 29 Sept. 2020, <https://doi.org/10.1038/s41598-020-72685-1>.
- Nurliana Nasution, et al. "Predicting Heart Disease Using Machine Learning: An Evaluation of Logistic Regression, Random Forest, SVM, and KNN Models on the UCI Heart Disease Dataset." *IT JOURNAL RESEARCH and DEVELOPMENT*, vol. 9, no. 2, 24 Apr. 2025, pp. 140–150, <https://doi.org/10.25299/itjrd.2025.17941>.

Rezaee, Mehrdad, et al. "Development and Validation of Risk Prediction Models for Multiple Cardiovascular Diseases and Type 2 Diabetes." *PLOS ONE*, vol. 15, no. 7, 29 July 2020, p. e0235758, <https://doi.org/10.1371/journal.pone.0235758>. Accessed 20 Nov. 2020.

van Dam, Peter M. "The Role of Machine Learning in the Early Detection of Cardiovascular Disease in a Community Setting." *European Heart Journal - Digital Health*, vol. 2, no. 1, 5 Feb. 2021, pp. 135–136, <https://doi.org/10.1093/ehjdh/ztab015>. Accessed 7 Nov. 2021.

Weng, Stephen F., et al. "Can Machine-Learning Improve Cardiovascular Risk Prediction Using Routine Clinical Data?" *PLoS ONE*, vol. 12, no. 4, 4 Apr. 2017, p. e0174944, <https://doi.org/10.1371/journal.pone.0174944>.

World Health Organization. "Cardiovascular Diseases (CVDs)." *World Health Organization*, 31 July 2025, [www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](http://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)).